



OIL INDUSTRY SAFETY DIRECTORATE

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WORK PERMIT SYSTEM
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PREFACE

Indian petroleum industry is the energy lifeline of the nation, and its continuous performance is essential for sovereignty and prosperity of the country. As the industry essentially deals with inherently inflammable substances throughout its value chain – upstream, midstream, and downstream – Safety is of paramount importance to this industry as only safe performance at all times can ensure optimum return of these national assets and resources including sustainability.

To ensure proper implementation of the various aspects of safety in the oil & gas industry, the Government of India, vide Resolution No. R-13013/4/84-OR-1 dated 10th January 1986 set up a Safety Council at the apex, under the administrative control of the Ministry of Petroleum & Natural Gas (MoP&NG). To assist the Safety Council, a technical directorate, namely the Oil Industry Safety Directorate (OISD) under the aegis of MoP&NG, has been entrusted with the responsibility of formulating standards, overseeing its implementation through safety audits to enhance safety level in the oil & gas industry.

OISD has developed a rigorous, multi-layer, iterative and participative approach for development of standards. OISD develops and revises Standards, Guidelines & Recommended Practices for the oil and gas sector through a participative process involving all the stakeholders (including the public at large), drawing inputs from international standards and adapting them to Indian conditions by leveraging the experience of the constituents. These standards cover inbuilt design safety, asset integrity and best operating practices in the field of production, processing, storage and transport of petroleum.

The participative process followed in standard formulation has resulted in excellent level of compliance by the industry. It also goes to prove the old adage that self-regulation is the best regulation. The quality and relevance of OISD standards had been further endorsed by their adoption/ reference in various statutory rules and regulations of the land.

OISD and industry together strive to achieve nil incident in the entire hydrocarbon value chain. We at OISD, are confident that the provisions of this standard, when implemented in totality, would go a long way in ensuring safe operation in the oil and gas industry.

**Executive Director
Oil Industry Safety Directorate**

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Wherever Acts/ Rules/ Regulation and National/ International Standards are mentioned in the standard, the same relates to in-vogue version of such documents. Whenever there is any difference between the statutory norm and OISD standard, more stringent requirement should be followed. However, in case of any difference in the requirement between OISD standards, the requirement as specified in the standard released at a later stage shall prevail.

The industry has to carry out gap analysis and make necessary changes in their system in line with the revised standard and the requirement should be implemented within a time frame of 1 year (from the date of release of revised standard).

Though every effort has been made to assure the accuracy and reliability of the information contained in these documents, OISD hereby expressly disclaims any liability or responsibility for loss or damage resulting from the use of OISD standard.

The standards would be reviewed periodically based on the decision in the Steering Committee/ Safety Council, industry request, technological development, recommendations of high-level committee and accident investigation reports. Generally, OISD standards are reviewed and revised, reaffirmed or withdrawn at least every ten years.

Suggestions for revision are invited from the oil and gas industry and should be submitted to OISD at **ps-ed.oisd@gov.in**.

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WORK PERMIT SYSTEM

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1.0 INTRODUCTION

The hydrocarbon processing/ handling installations store and process large quantities of hazardous substances, including flammable and toxic materials. Given the oil and gas industries' inherent hazards, most operating activities are critical to safety. An essential part of controlling workplace hazards is an effective Work permit System to prevent incidents, provide safe working conditions and execute the work safely.

The primary purpose of the work permit system is to ensure that work is carried out in the safest possible manner to prevent injuries to personnel, protect property/ environment from damage and avoid fire/ explosion. Failure to properly implement the work permit system is often the root cause of the incidents.

A work permit is not simply a means of permitting one to do a job. It is a means of defining how a job can be carried out safely and under what circumstances. Issuing a work permit does not, by itself, make a job safe. The work permit does not relieve anyone of responsibility for taking all necessary precautions to eliminate the possibility of accidents such as fire or explosion before, during, and after work is performed.

The words 'permit-to-work' (PTW) and 'work permit system' (WPS) are synonymous with the purpose of this standard and are used interchangeably in this standard. Similarly, the words 'Work', 'Job,' or 'Task' are commonly used interchangeably in this standard.

2.0 SCOPE

This standard shall apply to all upstream, midstream, and downstream sectors of the Oil and Gas Industry. The provisions of this standard are applicable for construction activities (like excavation, scaffolding, road work, working in confined space and at height, painting, electrical jobs, cutting/ welding, radiography, etc.) in Grass root/ Green field project sites.

3.0 DEFINITIONS

- 3.1. Authorized person:** is a person who possesses a certificate of competency or electrical work permit, issued by the appropriate Government required as per Central Electricity Authority (Measures relating to Safety and Electric Supply) Regulations, 2010 & its amendments thereof. This person can operate and carry out the work on electrical lines and apparatus.
- 3.2. Cold Work:** An activity which does not produce adequate heat to ignite a flammable air - hydrocarbon mixture or a flammable substance.
- 3.3. Confined space:** It is an enclosure that:-
 - (i) has limited openings for entry and exit.
 - (ii) could contain known or potential hazards.
 - (iii) is not intended for continuous occupation.
 - (iv) has insufficient natural ventilation.
- 3.4. Hot Work:** An activity that can produce a spark or flame or other sources of ignition having sufficient energy to cause ignition, where the potential for flammable vapour, gas, or dust exists.

In Petroleum Rules 2002, hot work is defined as any work which involves welding, burning, soldering, brazing, sand blasting, chipping by spark-producing tools, use of certain power driven tools, non-flame proof electrical equipment or equipment with internal combustion engines and including any other work which is likely to produce sufficient heat, capable of igniting inflammable gases.
- 3.5. Issuer:** A designated person authorized to issue the work permit by an organization and statutory guidelines. The issuer shall have the competency defined by the organization for executing the roles and responsibilities in the work permit system.



The list of the designated permit issuers for a plant/ location/ unit/ facility shall be maintained by the organization.

- 3.6. Lower Flammable Limit (LFL):** the minimum concentration of a vapour in the air (or another oxidant) below which flame propagation does not occur in contact with an ignition source. This is usually expressed as the volume percentage of the vapour in the air. This is also referred to as Lower Explosive Limit (LEL).
- 3.7. Personal Protective Equipment:** Devices or clothing used to insulate an individual from the chemical, thermal, explosive, electric shock, or other hazards presented by the environment in which one works. Examples of PPEs are safety spectacles/ goggles, safety shoes, safety helmets, cotton coveralls, hand gloves, ear muffs/ plugs, etc.
- 3.8. Pyrophoric substance:** A substance capable of self-ignition on short exposure to air under ordinary atmospheric conditions.
- 3.9. Receiver:** A designated person authorized to receive a work permit by an organization and statutory guidelines. He is responsible for the safe execution of the work. The receiver shall have the competency defined by the organization for executing his roles and responsibilities in the work permit system.

The list of the designated permit receivers for a plant/ location/ unit/ facility shall be maintained by the organization.

- 3.10. Shall:** 'Shall' indicates mandatory requirement.
- 3.11. Should:** 'Should' indicates that the provision is recommended as a good engineering practice.
- 3.12. Hazardous Area:** An area shall be deemed to be a hazardous area, where:-
- (i) Petroleum having a flash point below 65°C or any flammable gas or vapour in a concentration capable of ignition is likely to be present.
 - (ii) Petroleum or any flammable liquid having a flash point above 65°C is likely to be refined, blended, handled, or stored at or above its flash point.
- 3.13. Simultaneous Operations (SIMOPS):** Also called 'combined operations,' are those where more than one major activity (e.g., drilling, production, workover operation, revamping/ construction, line break containing flammable material and hot work in the area around etc.) is undertaken at the same place or in the vicinity along-with activities such as material handling and hot work in an environment where oil/ gas is present or produced nearby.
- Such 'combined operations' may conflict with each other and pose risks that may adversely impact people, property, or the environment.

4.0 OBJECTIVE & FUNCTION OF WORK PERMIT SYSTEM

The objective and function of a work permit system are as follows:

- (i) It ensures that proper authorization has been granted to perform designated work.
- (ii) It ensures those people conducting the work know the exact nature of the task, including hazards, restrictions, required precautions/ controls, time limitations, etc.
- (iii) It specifies the controls and precautions necessary to undertake the work safely, e.g., isolating machinery, equipment, pipeline, energy source, etc.
- (iv) It allows the ability to display, to those who need to know, exactly what ongoing work is.
- (v) It provides a procedural means of suspending work when this is necessary.
- (vi) It provides an ability to control work that might interact or conflict with ongoing operations or other work permit activities.
- (vii) It provides a procedural means of handing over the work when it covers more than one shift.
- (viii) It provides a procedural means of handing back the area or plant involved in the work.



5.0 TYPES OF WORK PERMITS

The organization shall define the list of jobs exempted from the work permit system. Such activities involving routine production and process operations. The organization shall develop and implement procedures for executing such jobs safely.

The work permit is required for jobs like various construction activities, inspection, repair and maintenance work, entry into confined spaces, work at height, excavation, pressure testing of pipelines, use of electrical equipment that is not recommended for hazardous areas, and entry/operation of vehicles that is not recommended for hazardous areas.

Based on the nature of the work to be performed, the following minimum type of work permits/ certificates shall be used:

- | | | |
|---|---|--|
| <ul style="list-style-type: none"> (1) Cold Work Permit (2) Hot Work Permit (3) Confined Space Entry (4) Electrical isolation and Energization permit | } | OR
Composite
Permit |
|---|---|--|

Composite Integrated permit for cold/ hot/ entry to confined space, work at height, radiography, excavation, vehicle entry, etc., may be used. However, it shall comply with all the requirements of individual permits.

5.1 COLD WORK PERMIT

Work falling under the category of cold work such as opening process machinery, blinding & de-blinding, tightening of flanges, hot bolting, inspection, painting etc. shall be performed through Cold Work Permit. Cold work permits cover a variety of potentially hazardous activities that do not involve "hot work" or are not of a type covered by a hot work permit. Cold work permits are issued when there is no credible ignition source and when all contact with harmful substances has been eliminated, or appropriate precautions are taken.

5.2 HOT WORK PERMIT

Hot work permits are applied to any work that involves actual or potential ignition sources and is done in an area where there may be a risk of fire or explosion or involves the emission of toxic fumes from the application of heat.

They are typically used for tasks involving naked flames or tasks having spark potentials like welding or flame/ gas cutting, burning, shot blasting, soldering, hot tapping on hydrocarbon pipelines or piping, metal drilling, use of any tools which may produce sparks (spark-producing grinding wheels, pneumatic hammers, and chippers, excavators for excavation) and use of electrical equipment, machine/ tool which is not intrinsically safe or is of non-flameproof proof type.

Entry and operation of vehicles inside hazardous areas (like for construction activities) also fall in the category of hot work and shall be performed under the hot work permit. Vehicles whose design (like engine exhaust system) is not approved by PESO and are used inside hazardous areas also require hot work permit. Alternatively, Organizations may decide to have separate permits for vehicle entry/ movement.

Some cold work activities might include a line break or blinding/ de-blinding on pyrophoric service, which contains fire hazards due to entrapped residual traces of the pyrophoric substance. Such activities shall be carried out through hot work permits or line break permits.

5.3 CONFINED SPACE ENTRY PERMIT

This permit is required to protect personnel entering a confined space such as a vessel, boiler, storage tank, large diameter piping, aircraft Refueler tank, etc., against hazards such as oxygen deficiency, toxic and flammable material, falling objects, power-driven equipment, etc. Excavation more than 1.2 meters deep, an entry on a floating roof tank when the roof is more than 3 meters down from the top, entry into internal floating roof tank, space located below ground level such as a pit, drain channel, etc., also falls under the confined space.

A confined space entry permit specifies the precautions to eliminate exposure to dangerous fumes, cave-in/ slides, or an oxygen-depleted atmosphere before a person is permitted to



enter a confined space. e.g., by maintaining forced ventilation and physical isolation or by providing personal protective equipment, including breathing apparatus. This permit shall confirm that the confined space is free from dangerous fumes or asphyxiating gases.

A confined Space Entry Permit shall be supplemented with a hot or cold work permit as per the type/ nature of work (if composite work permit is not used).

5.4 COMPOSITE WORK PERMIT

A composite work permit may be used for various jobs, such as cold/ hot work outside the confined space, entry to a confined space, and cold/ hot work inside the confined space. The checklist of the composite permit must address all the relevant aspects.

A typical composite work permit form for cold work, hot work, entry to confined space, vehicle entry, and excavation works is enclosed as Annexure-I. The explanatory notes of the composite work permit form are enclosed as Annexure-II.

Similarly, organizations may develop composite work permit forms for any combination of jobs. Checklist points of the composite permit may be sectionalized under various heads as given in the specimen copy. Based on the work to be performed, the relevant section checklist and the general section shall be duly ticked.

5.5 ELECTRICAL ISOLATION AND ENERGIZATION PERMIT

- (i) Before issuing any work permit, the equipment/ facility to be worked on must be electrically safe, and electrical power shall be isolated by an authorized person to the extent necessary for the safe conduct of the authorized work.
- (ii) All work on electrical equipment/ facility shall be carried out using electrical isolation and energization work permit.
- (iii) Permits for electrical isolation and energization shall be in two sections. Section-A shall be used for electrical isolation and Section B for energization. The typical format for the electrical isolation and energization work permit is enclosed as Annexure-III.

5.6 OTHER TYPES OF WORK PERMIT

The following activities/ jobs shall also be covered under the Work Permit system. Organizations should develop specific work permits for such activities or include them in the composite work permit:-

- (i) Working at heights
- (ii) Heavy lifts
- (iii) Excavation
- (iv) Line-break
- (v) Road closure
- (vi) Diving
- (vii) Radiography/ Work on the radiation source
- (viii) High-pressure water cleaning
- (ix) Quick sealing & hot tapping

6.0 WORK PERMIT SYSTEM PROCEDURES/ REQUIREMENTS

The term 'Work Permit System, 'permit,' or 'work permit' refers to the paper or electronic form which is used as part of an overall system of work and which has been devised by the organization to meet its specific needs.

The permits can be written, printed, or electronic, whichever meets the organization's needs. The organizations shall implement the requirements of this standard through an electronic or manual system. The principles and content of the paper permits shall be adopted in the electronic system. Digital signature of the electronic work permit forms is permitted.

The typical process flow chart of the work permit system is enclosed as Annexure-IV. The following are the requirements and procedures which shall be adopted for the implementation of the work permit system:-



6.1 ROLES & RESPONSIBILITIES

- 6.1.1 Organizations shall develop work practices/ procedures for the safe execution of jobs (like hot work, cold work, working in confined space, vehicle entry, working at heights, excavation, road closure, handling & lifting equipment, radiography, pressure testing, electrical, demolition, shot blasting/ spray painting, etc.).
- 6.1.2 Typical requirements to be included in safe work practices have been addressed in Annexure-V of this document. These practices shall address the requirements of certificates/ checklists required to support a work permit system such that the tasks are executed safely.
- 6.1.3 The Occupier/ Factory Manager/ Installation Manager (designated by the statutory laws) has overall responsibility for the development and implementation of the Work permit system, including defining roles and responsibilities of the Issuer and Receiver of the work permit system in line with the requirements of this standard.
- 6.1.4 The concerned management shall issue the appropriate authority levels for types of permits to the role holders at various installations. Alternate role holders shall be identified such that the work permit system is implemented correctly during the absence of the primary role holders from the location. The alternates shall be equally competent and trained to take responsibility.
- 6.1.5 The list of authorized personnel involved in the work permit system with their roles and responsibilities shall be available at the location. It shall be revised as and when necessary.
- 6.1.6 No hot/ cold job shall be undertaken without a work permit except in the areas pre-determined and documented in the organization's work permit system.
- 6.1.7 A copy of the organization's work permit procedures defined in line with this standard shall be available with all Installations/ plants.
- 6.1.8 The typical responsibilities of the permit role holders are given in Annexure VI.

6.2 JOB PLANNING

Job planning shall address the availability of authorized persons required to carry out safe isolations, handover, and execution of the job till the equipment is back in service.

6.3 JOB SAFETY ANALYSIS, CONTROL OF HAZARDOUS ENERGY SOURCES & TOOLBOX TALK

6.3.1 JOB SAFETY ANALYSIS (JSA)

- (i) For jobs for which SOP is unavailable (for safe execution), it is mandatory to carry out JSA before issuing the work permit.
- (ii) A careful Job Safety Analysis shall be made of the potential hazards and the operations to determine the appropriate safeguards before starting work.
- (iii) A job safety analysis process typically includes the following:-
 - description of a job.
 - breaking the job down into a sequence of steps to accomplish the task.
 - identifying the hazards, potential consequences, and risks associated with each of these steps.
 - determining the control measures (as per the hierarchy of controls/ risk-reducing measures) to overcome these hazards.
 - defining person responsible for implementing the control measures.
- (iv) In listing the control measures, general statements such as 'be careful, 'use caution,' etc. shall be avoided. Specific statements describing what action(s) is to be taken and how it can be performed shall be preferred.
- (v) JSA shall be work and job location specific. It shall properly document the safe precautions for carrying out the work applicable to the location site where work is carried out.



- (vi) Controls may include but are not limited to tools and equipment to be used, supervision and assistance required, work permits, safety observers, emergency response measures, and personal protective equipment. The action to be taken in an emergency shall be specified.
- (vii) Other activities (simultaneous operations) taking place or planned to occur simultaneously as the task to be performed shall be identified as they could impact the JSA.
- (viii) The JSA shall prescribe the frequency of the gas test (continuous detection using portable gas detectors or hourly or after every shift) & toolbox talks/ meetings. It shall also specify the site inspection process (who and when) during the execution of the work.
- (ix) The emergency response & evacuation procedure at the site shall be referred in the JSA. The workers shall know what to do in the event of an emergency.
- (x) If the JSA identifies hazards that cannot be addressed or if the proposed task poses unacceptable risks for people at the site, the work shall not be permitted.
- (xi) When the task cannot be carried out because of insufficient safeguards or SIMOPS, the task shall be immediately discarded or delayed until other operations due to take place at the same time have been completed.
- (xii) JSA should typically be prepared by a multidisciplinary team of relevant functions involved in task execution (like personnel from Fire & Safety depending on the critical nature of the job, M&I, Technical Services, Operations, and Executing Department. Wherever necessary, contractor/ his authorized representative should also be involved in the preparation of JSA. A typical JSA format is attached as Annexure-VII.
- (xiii) There should be a system of approval of JSA defined in the organization's work permit system. The approved JSA shall be attached to the work permit.
- (xiv) The JSA shall be reviewed and updated whenever a step changes or a new step is introduced in the job. The previous JSA shall be reviewed to ensure that the identified hazards and controls are relevant to the site, specific job, and personnel involved. Where appropriate, additional specific controls are to be identified.
- (xv) The actions generated from the JSA shall be assigned to the respective functions to ensure that these are completed before issuing the permit.
- (xvi) On completing the task, it is essential to capture any lessons learned and make improvements next time. The JSA shall be reviewed in case of an accident, incident, or near miss.

6.3.2 CONTROL OF HAZARDOUS ENERGY SOURCES

- (i) Organizations shall develop a system for physical lock-out tag out (LOTO) of Isolation based on the energy source. The lock-out and tag-out system shall be used to disable machinery or equipment, thereby preventing the release of hazardous energy while employees perform servicing and maintenance activities. It controls hazardous energies - electrical, mechanical, hydraulic, pneumatic, chemical, thermal, and other energy sources. Annexure-VIII outlines the lock-out and tagging procedure used as safe work practice when employees are exposed to a hazardous energy source.
- (ii) The most critical requirement of the lock-out and tag-out system is as follows: -
 - use lockout devices for equipment that can be locked out. Tag-out devices may be used instead of lockout devices only if the tag-out program provides employee protection equivalent to that provided through a lockout program.
 - ensure that new or overhauled equipment is capable of being locked out.
 - use only lockout/tag-out devices authorized for the particular equipment or machinery and ensure that they are durable, standardized, and substantial.
 - ensure lockout/tag-out devices identify the individual users.



- the receiver shall also ensure that equipment is isolated before performing the job.
- (iii) The isolations appropriate to the job shall be specified on the permit/ supplementary isolation certificate and correctly implemented.
- (iv) All necessary people shall be aware of the isolated equipment.

6.3.3 TOOLBOX TALK

- (i) Toolbox talks (or meetings) are intended to facilitate health and safety discussions on the job site and promote safety culture. It is aimed to refresh workers' knowledge, cover last minute safety checks, to raise workers' awareness of specific problems at job site.
- (ii) Toolbox talk (TBT) shall be conducted at the job site before the commencement of the job, and start of the subsequent shifts/ daily in case the work is extended. The frequency of the tool box meetings depends upon the risks associated with the job to be carried out. The frequency of tool box meetings shall be documented in the JSA. TBT should be repeated if there is any change in the personnel executing the job.
- (iii) It shall include all people involved in the work or those affected, such as subcontractors, vendors, contract supervisors, and contract workers. The job supervisor/ engineer-in-charge and safety officers should also attend these meetings based on their availability and criticality of the jobs.
- (iv) The permit receiver/ job supervisor shall impart it.
- (v) It shall cover safety topics related to the specific job, such as workplace hazards, emergency procedures, do's & don'ts, and safe work practices. The JSA shall be discussed during the toolbox talk. It shall involve the review of the potential hazards identified, and control measures (like PPE to be used and safety procedures) put in place to mitigate the hazards. Toolbox Talks should be used for post-accident communications/ accident case study, and pre-task planning etc.
- (vi) Records of such talks should be maintained. The typical format of the toolbox talk form is attached as Annexure-IX.
- (vii) Following information should be included when completing a Toolbox Talk Form:
 - Names and signatures of all persons present.
 - Topics discussed.
 - Any job specific training or instructions that have been given in the meeting and who raised the issue or added to the discussion, to show that there has been active consultation with all persons attending the meeting and that all parties have had a chance to raise issues.

6.4 WORK PERMIT SYSTEM ACTIVITIES BEFORE COMMENCEMENT OF JOB

6.4.1 ESSENTIAL REQUIREMENTS FOR ISSUING & RECEIVING WORK PERMITS

- (i) Work shall not start without a valid work permit.
- (ii) While working at a height of more than 1.8 meters, working at height permit shall be issued before commencement of the job.
- (iii) The single permit shall not be issued to multiple agencies.
- (iv) The work permit shall be issued on the day of the commencement of the job.
- (v) All work permits shall be issued or received by the authorized person of the area where work is to be carried out.
- (vi) The work permit shall not be issued and received by the same authorized person, i.e., a person shall not issue a permit to himself. However, the issuer and receiver can be from the same Department.



- (vii) The work may be planned to be done either departmentally or through a contractor. In either case, the work permit preferably be received by the organization's authorized person. However, in some instances (like single-man locations), the permit may be directly received by the contractor's authorized supervisor. Such exceptions shall be documented in the organization's work permit system.
- (viii) The permit-issuing authority shall not be delegated to the contractor/ his representative except in the following situations: -
- Retail Outlet, if not directly run by the organization: The permit issuing authority can be delegated to the Dealer/ Manager of the Retail Outlet, provided they are competent and authorized by the organization.
 - Entire operations are outsourced on a contract/ turnkey project basis: the work permit system shall be governed by the approved 'bridging document.' During the preparation of contract agreements with outsourcing parties like in Rigs and Installations, the bridging document shall confirm that the implemented work permit system meets this standard's requirements.

6.4.2 WRITING WORK PERMIT FORMS

- (i) The work permit shall be filled up carefully and accurately in clear writing, ensuring detailed and complete information is provided in all the sections/ subsections. The relevant sections and conditions of the permits shall be appropriately ticked in the permit form. Poor communication can cause mistakes and accidents.
- (ii) The permit form shall have sufficient space against 'Description of work' to be done to give a clear and complete description of the work.
- (iii) Other factors to be ensured while filling up the work permit form are: -
- (1) sentences shall be short and simple while writing a description of work, additional precautions, remarks, etc., in the permit form.
 - (2) exact nature and scope of the work shall be clearly described.
 - (3) the details of the personnel (name and designation) issued to and involved in the permit shall be mentioned, like Issuer and Receiver.
 - (4) exact plant or geographical location of the work shall be specified.
 - (5) equipment details to be specified.
 - (6) the date and time that work can start.
 - (7) the period of the time the permit is valid for (operational time limit of the permit).
 - (8) details isolation certificates confirming implementation of the isolations appropriate for the job.
 - (9) cross-reference of common isolations.
 - (10) cross-reference of other relevant work permits.
 - (11) quantitative values and limits of gas test values shall be numerically specified.
 - (12) the applicable conditions/ checklist items of the permit shall be correctly ticked
 - (13) signatures on the work permit shall be dated.
 - (14) time of signature shall be mentioned.
 - (15) signatures and initials shall be pre-authorized by the organization, traceable and legible.
 - (16) electronic work permits can be digitally signed.
- (iv) Legible certificates and attachments (sketches/ drawings) shall be attached to the work permit (manual/ electronic) wherever required to avoid miscommunication.
- (v) Additions and alterations due to changes in identified conditions during the execution of the work shall be endorsed on all copies of the work permit.

6.4.3 PRINTING & SIGNING FILLED-IN WORK PERMIT FORMS

- (i) Copies of the work permit shall be distributed to appropriate personnel. Permits are often issued in triplicate. Wherever available, the emergency personnel (Fire, First aid, Rescue, etc.) shall be advised of the work and its exact location. The copy of the permit shall be available at the worksite either as a printed hard copy or viewable in electronic form (like in a handheld device suitable for the hazardous area).



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- (ii) All permits shall be serially numbered. General instructions applicable to the type of permit (manual/ electronic) shall be printed on the overleaf.
- (iii) In a manual system, work permit forms shall be in the form of books preferably with a tear-off facility. In the case of an electronic permit system, the permit shall be accessible to the relevant stakeholder. A copy of the work permit shall be printed in case the handheld devices are unavailable with the crew at the work site.
- (iv) Each organization shall develop a unique system to recognize different types of permits in manual or electronic work permit systems. The method may include color-coding the various permit forms or identifying the types of work permits by BOLD letters or symbols.
- (v) The suggested color-coding system & the distribution of the copies are as follows:-

	Type of work permit	Color code	No. of copies	Copies to be distributed to
(1)	Hot work permit	Pink-edged or pink	Three	• Original: with receiver/ acceptor • Copy: Fire/ Emergency Services • Book Copy: Control Room/ designated location
(2)	Cold work permit	Yellow-edged or yellow	Two	• Original: with receiver/ acceptor • Book Copy: Control Room/ designated location
(3)	Confined space entry work permit	Colour other than that of cold and hot work permit	Three	• Original: with receiver/ acceptor • Copy: Fire/ Emergency Services • Book Copy: Control Room/ designated location
(4)	Composite work permit	White	Three	• Original: with receiver/ acceptor • Copy: Fire/ Emergency Services (if applicable) • Book Copy: Control Room/ designated location
(5)	Electrical isolation/ energization permit/ certificate	Green-edged or green	Two	• Original, along with a copy, shall be issued to the electrical section for electrical isolation/ energization. • The electrical section authorized person on isolation/ energization of the equipment/ circuit shall return the original to the issuer keeping a copy for record.

- (vi) The plant in charge/ his authorized representative/ permit issuer shall make a thorough inspection of the job site, accompanied by a Fire/ Safety Officer (as per requirement), area in-charge and permit receiver.
- (vii) Clearance from the relevant authorities/ Department shall be obtained for excavation/ Road cutting /dyke cutting jobs.
- (viii) The work permit will become live only after signing by the Permit Issuer (who authorizes the permit as issued) and the Permit Receiver (who accepts the permit confirming understanding of the work to be done, hazards involved, precautions required, and that the permit information has been explained to all personnel executing the job). Their signatures formally confirm that all necessary precautions have been taken and that work can now start, providing the necessary precautions are adhered. Their name and signature and the date and time should be clear.
- (ix) The names and designation of the permit authorities (like issuer and receiver) shall be conspicuously visible in the printed work permit form.



6.4.4 WORK PERMIT INTERACTION

- (i) When a work permit is being considered for a job, the person issuing the permit must be aware of any other activity (either planned or already underway) which may influence the work to be done under the permit.
- (ii) In case multiple issuers exist at the workplace, close liaison is required by the various issuers to ensure that work done under one permit does not interact with another, creating hazards for either of them.
- (iii) All common isolations shall be cross-referenced in relevant work permits. In case of isolations common to more than one permit, there shall be a procedure to prevent the isolation from being removed before all permits have been closed.
- (iv) If more than one department works on the same equipment, a work permit shall be issued individually each department/ agency.

6.4.5 DISPLAY OF WORK PERMITS

- (i) Copies of a work permit shall be displayed/ available:
 - at the place where the work is being carried out or in a recognized close-by location.
 - the copy shall be available with the executor at the site.
- (ii) Copies of work permits issued for the same equipment/ facility shall be kept and displayed together.
- (iii) The entity should have the provision to display the real-time status of critical work permits (like hot work, working at height, excavation, entry inside confined space, radiography) on the entity plot plan either digitally or manually. This information shall be available to the critical stakeholders at the facility, role holders of the work permit system, fire stations, and unit control rooms. This helps the incoming shift supervisor/s (both in fire stations and operation departments) get a quick idea of the hot jobs being undertaken and helps identify the areas that require inspection/ attention, depending upon the criticality of the area and the job. The site of hot jobs shall be indicated in this system (like using red pins on manual plot plans).

6.5 WORK PERMIT SYSTEM ACTIVITIES WHILE EXECUTING JOB

6.5.1 PERMIT SUPERVISION

- (i) The area in charge shall be aware about the commencement of work. Once the job is commenced, it is essential to monitor the worksite for any changes in conditions or personnel that might alter the hazards and controls in place.
- (ii) Site visits by the issuing authority, area in-charge, authorized person & safety officer/ fire officer (depending on the requirement) will depend on the complexity and duration of that work, and the hazards involved. In the case of hot work and confined space jobs, the site visits shall be carried out at least twice – at the start and completion of the work but more often if required. They shall verify whether the work is being carried out as per the conditions laid down in the work permit. The housekeeping of the site shall be proper to avoid incidents while executing job.
- (iii) Suitable technologies like CCTV and gas monitoring devices should be utilized to monitor site conditions continuously when work is being carried out. Signs and barriers shall be used in the work area to prevent the entry of unauthorized persons, pedestrians, and vehicles.
- (iv) Adequate supervision of the work permit system might get diluted in the case of many permits under one person's control. Hence, management shall ensure the necessary arrangements to maintain proper supervision of the jobs executed through work permit.



6.5.2 SUSPENSION & CANCELLATION OF WORK PERMIT

- (i) Many incidents happen when conditions at the worksite change or are not as foreseen. Work shall be stopped if any unsafe condition arises during work and the permit authorities shall be notified immediately.
- (ii) Work being executed under a work permit should also be suspended:-
 - if there is an emergency alarm/ siren.
 - for operational reasons.
 - if site conditions change or new hazards have arisen
 - there is a change to the nature or scope of the work.
 - where the ongoing work conflicts with other work being carried out.
- (iii) The permits which have been suspended shall still be kept on the permit recording system. They shall specify the condition of the plant where the work has been suspended and any consequences for other activities which may be affected by that condition.
- (iv) Isolations for the suspended work permits shall not be normalized till the work permit is closed.
- (v) The conditions shall be restored before restarting the work. The suspended work shall not be restarted until the permit issuer has verified that it is safe to do so and has revalidated the risk assessment and corresponding work permit.
- (vi) If the suspension of work is likely to be extensive/ indefinite, then the work permit shall be cancelled, and a new work permit with additional/ appropriate safeguards shall be issued to restart the work.

6.5.3 WORK PERMIT EXTENSION/ HANDOVER

- (i) The permit extension is required if any work done under a work permit carries over into another shift. This allows the outgoing shift to communicate all relevant information about the work and the conditions of the permit to the new shift to ensure continuity and safety.
- (ii) The information shall include the following: -
 - any outstanding work under permit control.
 - the status of that work (this shall be left in a condition that can be reliably explained to and understood by the incoming shift).
 - the status of any other work which may affect the permit-controlled job (e.g., isolations).
 - the status of the plant or installation.
- (iii) The information can be recorded in a permit log, permit file, or on display boards. It is essential, however, to have good communication between the outgoing and incoming issuing and performing authorities.
- (iv) The names and designation of the permit authorities, like issuer and receivers, shall be mentioned in the permit renewal sections.
- (v) The permit shall be issued only for a single shift.
- (vi) Where the work has to be continued, the same permit may be revalidated/ extended shift-wise for a period not exceeding seven calendar days in the succeeding shifts after satisfying the permit conditions.
- (vii) In plant turnaround instances, work permit validity may be extended to fifteen days.
- (viii) For construction activities to be undertaken in grass root project sites and turnkey projects, a Work Permit of validity not exceeding one month with periodic checks may be issued with the approval of the designated senior management. The frequency of periodic checks shall be mentioned in the Job Safety Analysis.



6.6 HANDBACK/ CLOSING OF WORK PERMIT AFTER EXECUTION OF JOB

- 6.6.1 The permit closing/ hand back procedure is reinstating a plant or installation (or the part that has been worked on under a permit). This will ensure that the work has been completed, the worksite is in a safe condition, and normal operations can resume.
- 6.6.2 After completion of the job, the work permit shall be immediately closed.
- 6.6.3 The receiver shall satisfy himself about the completion of the job (thoroughly check the area for clearing of debris, removal of temporary electrical installation & machine, etc.) before closing the permit.
- 6.6.4 Similarly, the issuer shall satisfy himself regarding completing the job and that the area has been cleared before closing the permit. He is responsible for signing off the permit and verifying that the plant or installation is safe.
- 6.6.5 The person in control of operational activities will then refer to the permit that the plant or equipment (or the part which has been subject to work under a permit) has been reinstated to the control of the production staff.
- 6.6.6 The permit will be treated as "Closed" only after it is signed by Issuer and the Receiver (whether manually or electronically).
- 6.6.7 Electrical equipment or any other energy source is to be re-energized/ normalized only after the work on the equipment/ circuit has been completed and all the applicable work permits are closed.

6.7 RETENTION OF WORK PERMIT FORMS

- 6.7.1 On completion of the work and closing of the work permit, the original work permit shall be returned to the issuer.
- 6.7.2 Original along with other copies of work permits, supplementary checklists/ certificates, JSA, and risk assessments shall be properly filed as per serial number and kept as records for future reference and to enable auditing or incident investigation for at least six months from the date of completion of work or the successful commissioning of the facility for which the work permit was issued whichever is later.
- 6.7.3 Companies shall have a secure system for archiving and retrieving completed/ used work permits in the electronic permit systems to enable an effective monitoring and audit process.

6.8 AUDITING AND REVIEW OF THE WORK PERMIT SYSTEM

- 6.8.1 The work permit system implemented at the locations shall be audited during both internal & external safety audits (like OISD). The audit process shall examine monitoring records. Non-compliance with the work permit system shall be recorded, and subsequent remedial measures tracked to ensure all issues are effectively closed out.
- 6.8.2 The site and corporate management shall review the organization's work permit system annually to assess its effectiveness. This review shall include specific incidents related to inadequate control of work activity. It shall consist of lessons learned from incidents associated with the work permit system, safety alerts and case studies issued by OISD, and findings from internal and external audits.

6.9 TRAINING ON THE WORK PERMIT SYSTEM

- 6.9.1 As part of a site or installation safety induction training (fresh entrants and transferred employees from other Departments/ locations/companies), training on the work permit system shall be imparted through the course modules prescribed in OISD-STD-154.
- 6.9.2 The organization shall also have a system for refresher training for its employees who are involved in the work permit system.
- 6.9.3 Since several maintenance/ construction jobs are often carried out with assistance from contractors, it is essential to provide sufficient exposure to the contractor and its employees as well on the work permit system.



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6.9.4 The following topics shall be covered in the training sessions:

- objectives & principles of the work permit system.
- when permits are required.
- an understanding of the types of permits.
- job safety analysis & tool box talk procedure.
- responsibilities and competence requirements for signatories or authorized people within the work permit system.
- lessons from incidents associated with permits-to-work and findings from audit and review.

6.9.5 Once training has been given, acknowledgment shall be taken from the participants about the understanding of the requirements of the work permit system.

6.9.6 Records of training and competence of personnel involved in the work permit system shall be maintained for a minimum of three years.

7.0 ABBREVIATIONS

CCTV	Closed-Circuit Television
ELCB	Electric Leakage Circuit Breaker
JSA	Job Safety Analysis
LFL	Lower Flammable Limit
LEL	Lower Explosive Limit
LOTO	Lock Out Tag Out
M&I	Maintenance & Inspection
PESO	Petroleum and Explosives Safety Organization
PPE	Personal Protective Equipment
SIMOPS	Simultaneous Operations
TBT	Tool Box Talk



8.0 REFERENCES

- (i) The Petroleum Rules, 2002.
- (ii) Oil Mines Regulations.
- (iii) Factories Act, 1948.
- (iv) API RP 2009, Safe Welding, Cutting, and Hot Work Practices in Petroleum and Petrochemical Industries.
- (v) API STD 2015 – 2001, Requirements for safe entry and cleaning of petroleum storage tanks.
- (vi) Task Risk Assessment, Step Change in Safety.
- (vii) Health & Safety Executive: HSG250, Guidance on permit-to-work systems.
- (viii) OISD-STD-137, Inspection of electrical equipment.
- (ix) OISD-STD-154, Safety aspects in functional training.
- (x) OISD-GDN-192, Safe Practices during construction.
- (xi) OISD-GDN-207, Contractor Safety.
- (xii) OISD-STD-190, Safety in Derrick Floor Operations (Onshore & Offshore Drilling Rigs).
- (xiii) IS 5216, Recommendations on Safety Procedures and Practices in Electrical Work.
- (xiv) IS 3016, Code of practice for fire precautions in welding and cutting operations.
- (xv) OSHA standards 1910.147, 1910.333 & 1910.146.



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ANNEXURE - I

TYPICAL COMPOSITE WORK PERMIT
FOR
COLD WORK/ HOT WORK/ ENTRY TO CONFINED SPACE/ VEHICLE ENTRY/
EXCAVATION WORKS

(NAME OF ORGANIZATION)
(LOCATION)

Sl. No. _____

Work clearance from _____ hrs. of date _____ To _____ hrs. of date _____
(Valid for the shift unless renewed)

Issued to (Department/ Section/ Contractor) _____

Exact Location of work (Area/ Unit/ Equipment no. etc.) _____

Description of work _____

JSA Ref. No. _____ Cross-Reference of other permits: _____

Ref. No. of isolation certificates _____

THE FOLLOWING ITEMS SHALL BE CHECKED BEFORE ISSUING THE PERMIT
(Tick mark in the appropriate box.)

Sr. no.	Item	Done	Not Reqd.	Sr. no.	Item	Done	Not Reqd.
A	General points and for cold work			14	Proper ventilation and lighting provided		
1	Equipment/ work area inspected			B	For Hot work/ Entry to confined Space		
2	Surrounding area checked, cleaned and covered			2	Proper means of exit/ escape provided		
3	Sewers, manholes, CBD etc. and hot surfaces nearby covered			3	Standby personnel provided from Process/ Maint./ Contractor/ Fire dept.		
4	Considered hazard from other operations and concerned persons alerted.			4	Checked for oil and gas trapped behind the lining in Equipment		
5	Equipment properly drained and depressurized			5	Shield provided against spark		
6	Equipment properly steamed/ purged			6	Portable equipment/ nozzles properly grounded		
7	Equipment blinded/ disconnected/ closed/ isolated/ wedge opened			7	Standby persons provided for entry to confined space		
8	Equipment water flushed						
9	Iron sulfide removed/ kept wet			C	For Vehicle Entry		
10	Equipment electrically isolated and tagged vide permit no.			1	PESO approved spark elimination system provided on the mobile		



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					equipment/ vehicle provided.		
11	Gas test: HCs = %LEL Toxic gas = ppm O2 = %						
12	Running water hose/ Fire extinguisher provided. Fire water system available.			D	For Excavation works		
13	Area cordoned off and precautionary tags/ boards provided.			1	Clearance obtained for excavation/ road cutting/ Dyke cutting from concerned Department and Regulatory Authorities.		

REMARKS:

- (1) The activity has the following expected residual hazards (Tick the relevant items): Lack of Oxygen/ H₂S, Toxic gases/ Combustible gases/ Pyrophoric Iron/ Corrosive Chemicals/ Steam – Condensate / Others
- (2) Following PPEs to be used in addition to standards PPEs (Helmet, Safety Shoes, Hand gloves, Boiler suit): Cotton Coverall/ Face Shield/ Apron/ Goggles/ Dust Respirator/ Fresh Air Mask/ Lifeline/ Safety Belt/ Airline/ Earmuff, etc.
- (3) Additional requirements/ precautions if any: _____
- (4) _____

Issuer Name & Designation	Issuer Signature with date and time	Receiver Name and Designation	Receiver Signature with date and time

Clearance renewal

Date	Time		Gas Test Values for HC's, Toxic, O ₂ , etc.	Additional precautions if any, Otherwise mention "NIL"	Issuer's Name, Designation & Signature	Receiver Name, Designation and Signature
	From	To				

Closing of the work permit:

Receiver: Certified that the subject work has been completed / stopped and area cleaned.			Issuer: Verified that the job has been completed and area cleaned and is safe from any hazard.		
Date & Time	Name Designation & Signature		Date & Time	Name Designation & Signature	

**General Instructions:**

1. The work permit shall be filled up carefully and accurately in clear handwriting ensuring that complete information is provided in all sections/ subsections and none of the columns is left blank. Sketches shall be provided wherever possible to avoid miscommunication.
2. Appropriate safeguards and required personal protective equipment shall be determined by a careful analysis of the potential hazards and the operations to be performed prior to starting the work.
3. Requirement of standby personnel from Process/ Maintenance/ Contractor/ Fire etc. if any shall be mentioned in the additional requirement.
4. There shall be means of communication in case of an emergency at the worksite. In case of fire alarm/ siren, all work must immediately be stopped.
5. Only certified vehicles/ engines and permitted types of electrical equipment and tools are allowed in operating areas.
6. Welding machines shall be located in ventilated areas.
7. No hot work shall be permitted unless the explosive meter reading is Zero.
8. When a person is entering the confined space, a minimum one standby-designated person shall always be available at the manhole or entry point. If a standby person has to enter the confined space, then a substitute standby person shall be ensured at the manhole or entry point.
9. Compressed gas cylinders shall not be allowed to be kept in confined spaces.
10. Before boxing up any vessel manhole cover, ensure that no men/ materials are inside the vessel.
11. For renewal of work clearance, the issuer shall ensure that the conditions are satisfactory for the work to continue. If the conditions have changed, it may be necessary to issue a new permit or amend the existing permit.
12. This clearance on the same permit can be renewed/ extended up to a maximum of seven calendar days.
13. This permit must be available at the work site at all times.
14. On completion of the work, the permit must be closed and kept as record.
13. The industry may add other relevant instructions based on their operating and maintenance practices.



Annexure - II

EXPLANATORY NOTES TO COMPOSITE WORK PERMIT FORM

Explanatory notes to fill up the work permit (Annexure-II) and its check-listed items are elaborated below to amplify the underlying concepts and highlight their significance:

- (a) **Exact Location (Area/ Unit/ Equipment no):** Exact location of the area or the unit in which the work is to be carried out shall be written against it. E.g. AVU/ 12C-101, Tank farm A, Tank 501 etc. Wherever the equipment/ area is not identified, it shall be identified with the nearest/ smallest distance from a nearby equipment/ structure.
- (b) **Description of work:** Precise description of the work to be performed shall be written. Sketches shall be attached wherever possible to avoid miscommunication.
- (c) **Equipment/ Area inspected:** Equipment or area where work is to be conducted, shall be inspected to ensure that it is safe to carry out the work and assess other safety requirements/ stipulations. In case of equipment/ vessel box-up permits, it shall be ensured that the work is complete, all personnel are out, no maintenance gear is left behind, and debris removed.
- (d) **Surrounding area checked/ cleaned:** Unsafe conditions for performance of work may arise from the surrounding area. It shall be cleaned up to remove flammable material such as oil, rags, grass, etc.
- (e) **Sewers, Manholes, Closed Blowdown (CBD) etc., and Hot Surfaces covered:** Flammable gases may be released from nearby sewers. Hot un-insulated surfaces (equipment/ pipelines) may provide a source of ignition, and therefore, these are to be adequately isolated/ covered to prevent fires.
- (f) **Hazard from other routines/ non-routine operations considered and persons alerted:** Other activities (routine/ non-routine) being carried out nearby, which can create conditions unsafe for performance of the permit work, shall be taken into consideration and the concerned persons shall be alerted accordingly.
- (g) **Equipment electrically isolated and tagged:** Before issuing a permit, it shall be ensured that electrical isolation has been done, switches are locked-out and cautionary tags duly signed with date and time are attached. Single line drawing with isolation points shall be attached.
- (h) **Running water hose/ Portable extinguisher provided/ Fire water system available:** Running water hose and portable fire extinguisher are required to flush/ dilute in case of release of any hazardous chemical or to quench sparks and to put out small fires immediately.

In order to meet any contingency, it shall be ensured that the fire water system including firewater pumps, storage, network etc. is checked and kept ready for immediate use.

For the open flame hot jobs, continuous gas monitoring shall be done.
- (i) **Equipment blinded/ disconnected/ closed/ isolated/ wedge opened:** Equipment for which the work permit is being issued, shall be isolated from the rest of the plant in order to ensure that there is no change in the work environment with respect to presence of toxic/ flammable gases, liquids, hazardous chemicals etc. in the course of the work. The equipment shall be made hydrocarbon/ energy free.

Blinding is one of the most effective ways of isolation. Blinds shall be installed as close to the equipment as possible. If lines cannot be blinded, these shall be disconnected and the open ends shall be made safe by installing pipe caps/ plugs, blind flanges etc.
- (j) **Equipment properly drained/ depressurized:** Equipment under pressure shall be depressurized after isolation. This shall be followed by draining/ purging/ water flushing etc. as the case may be.

Equipment containing liquid hydrocarbons shall be drained completely. There may be a possibility of overlooking of liquid collected in pockets or inaccessible areas such as level gauges, small nozzles / bleeders on vessels, laterals in pipe work etc.

All low point drains shall be in unplugged condition. Before any hot job in a petroleum storage tank, it has to be certified to be free of hydrocarbon gases.



- (k) Equipment properly steamed/ purged:** Purging of equipment (tanks, vessels, pipelines etc.) is done to make them free of flammable hydrocarbon and toxic gases. Steam/ Inert Gas is used for gas freeing of vessels and pipes in refineries and other locations. Other means of purging is by displacement with water and final traces of gas removed by a properly earthed air eductor. All high point vents shall be unplugged while purging. It shall be done in a systematic manner to cover the entire equipment/ plant and continue till the concentration of toxic/ flammable gas is lowered to allowable levels.
- (l) Equipment water flushed:** Water flushing is an effective means of cooling, cleaning and even gas freeing of equipment. It is also employed to remove traces of acids/ chemicals. Equipment metallurgy must be considered before using sea/ saline water. Sometimes, flushing with demineralized water would be necessary depending upon the metallurgy of the equipment.
- (m) Gas/ Oxygen deficiency test done:** Gas test includes measurement of:
- (i) Hydrocarbons.
 - (ii) % Oxygen concentration.
 - (iii) Toxic gases like Hydrogen Sulphide, Sulphur dioxide, Carbon Monoxide, and Chlorine etc.
- The gas test shall be done using a calibrated instrument by trained gas testers. The calibration records of the instruments shall be maintained. Minimum two instruments should be available.
- (n) Shield against sparks provided:** Hot works like cutting, welding, grinding, etc. generates sparks which can provide a source of ignition to the surroundings. In order to protect the operating area from the hazards of sparks generated, shields are to be provided to contain the sparks generated. The shield material shall be non-flammable, shall not contain asbestos and shall be kept wet with water.
- (o) Proper ventilation and lighting provided:** Where natural ventilation is not available, fans/ air eductors shall be provided. These are also required for speedy dispersal of fumes generated by welding jobs. Only approved reduced voltage extension lights (not exceeding 24 volts) conforming to the hazardous area classification are to be allowed for work inside vessels.
- (p) Proper means of exit provided:** Proper means of exit is required in case of emergencies developed on account of the work or otherwise. Availability of an alternate route of escape shall be considered.
- (q) Area cordoned off and caution boards provided:** To prevent any unwarranted entry in the work area and also to caution other personnel taking actions which may endanger people working on the permit job, precautionary tags/ boards are to be provided to display like "No Entry" sign on roads or "Caution - Men at Work Inside" on the manhole of a vessel, "Danger - Radiography in progress" etc.
- (r) Portable equipment/ Hose nozzles properly grounded:** As a precaution against static electricity generation, portable equipment/ hose nozzles e.g. nozzle of a shot blasting gun, are to be grounded. Use of hydrocarbon lines for earthing shall be avoided. 30 milliamp earth leakage circuit breaker (ELCB)/ residual current circuit breaker (RCCB) shall be installed on electrical circuits/ cables feeding power supply to portable equipment.
- (s) Standby person provided for entry to confined space:** Whenever a person is entering in a confined space, minimum one designated person shall be kept at the manhole or entry point. The designated persons shall be in constant communication with the persons inside the confined space. The designated person's name shall be mentioned in the permit during revalidation in each shift. When a person is entering the confined space, a minimum one standby-designated person shall always be available at the manhole or entry point. If a standby person has to enter the confined space, then a substitute standby person shall be ensured at the manhole or entry point.
- (t) Standby personnel provided for fire watch from Process/ Maintenance/ Contractor/ Fire Department:** Depending on the criticality of the job, work permit issuer shall decide the type of standby to be provided i.e. from which department, of which level, how many and also additional firefighting support facilities etc.
- (u) Pyrophoric substance removed/ kept wet:** Pyrophoric substances may be present in the operating area/ equipment handling hydrocarbons. Iron Sulphide scale is the most common



pyrophoric substance encountered. These shall be either removed to safe locations or kept wet all the time to prevent their auto-ignition.

- (v) **Clearance obtained for excavation/ Road cutting/ dyke cutting etc.:** For any excavation work which may affect underground sewers/ telephone lines/ cables/ pipelines etc., clearance shall be obtained from all the concerned sections. Markers shall be put around the area where excavation is to be done and the depth to be indicated in the work permit.

Road cutting can hamper the movement of the fire vehicles; initial clearance shall be obtained from the Fire Department and final approval from the higher designated authorities.

When the dyke is cut, any mishap in the tank farm can lead to a free flow of oil to outside the bund. A higher level authority shall be designated for authorizing dyke cutting. Further, it shall be ensured that the dyke would be reconstructed in the shortest possible time.

- (w) **Checked spark arrestor on mobile equipment:** No vehicle/ engine without PESO approved spark arrestor or spark elimination system shall be permitted in operating areas. However, if the exhaust of the diesel engine is based on design having electronic fuel management with unit injectors and electronic control unit coupled with turbo charger and intercooler arrangements, no separate spark arrestor need to be provided. However such models shall have certification from OEM and PESO approval.

- (x) **Checked for oil/ gas trapped behind lining in equipment:** Before undertaking hot jobs, a check shall be done for oil/ gas trapped behind lining in the equipment.



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ANNEXURE - III

TYPICAL ELECTRICAL ISOLATION/ ENERGIZATION PERMIT

(NAME OF ORGANIZATION)

(LOCATION)

SECTION-A: ISOLATION PERMIT.

Permit no.: _____ Cross-Reference of other Permit: _____

Request for Isolation:

Date: _____ Time: _____

Department/ Section/ Area issuing the permit _____

Equipment number to be isolated: _____

Name of the equipment/ circuit to be isolated: _____

Description of the work to be carried out: _____

The above-mentioned equipment/ circuit shall be de-energized and isolated from all live conductors to carry out the maintenance work by _____ section/ for operational requirement.

Issuer Name_____
Designation_____
Signature**Certificate of Isolation:**

Date: _____ Time: _____

Certified that Equipment/ Circuit no. _____ of _____ plant has been made dead, is electrically isolated from all live conductors and has been connected to earth and the work mentioned above can now be carried out in accordance with the safety rules and regulations. For the purpose of making the equipment dead, the switches/ isolators/ links/ fuses (tick as applicable), have been opened and the section so isolated has been earthed at each isolation point and danger notice plates fixed thereon. The LOTO tag/ device no. _____ is put on the supply panel. Actions in respect of electrical isolation have been recorded in the electrical shift logbook including the instructions for the person who may relieve me.

Name of Authorized Person_____
Designation_____
Signature

.....



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SECTION-B: ENERGIZATION PERMIT.

Permit no.: _____

Request for Energization:

Date: _____ Time: _____

Department/ Section/ Area issuing the permit _____

Equipment number to be energized: _____

Name of the equipment/ circuit to be energized: _____

The above mentioned work on the subject equipment/ circuit has been completed and all the applicable work permits are closed. This equipment/ circuit may be energized.

_____ Issuer Name	_____ Designation	_____ Signature
-----------------------------	-----------------------------	---------------------------

Certificate of Energization:

Date: _____ Time: _____

Certified that Equipment / circuit no. _____ of _____ plant has been electrically energized by closing the switches/ isolators/ links/ fuses (tick as applicable), and the equipment put back into service.

The danger LOTO tag/device no. _____ is removed from the supply panel.

The temporary earths and other connections and danger notice plates made have been removed.

Necessary entry is also recorded in the electrical shift logbook.

_____ Name of Authorized Person	_____ Designation	_____ Signature
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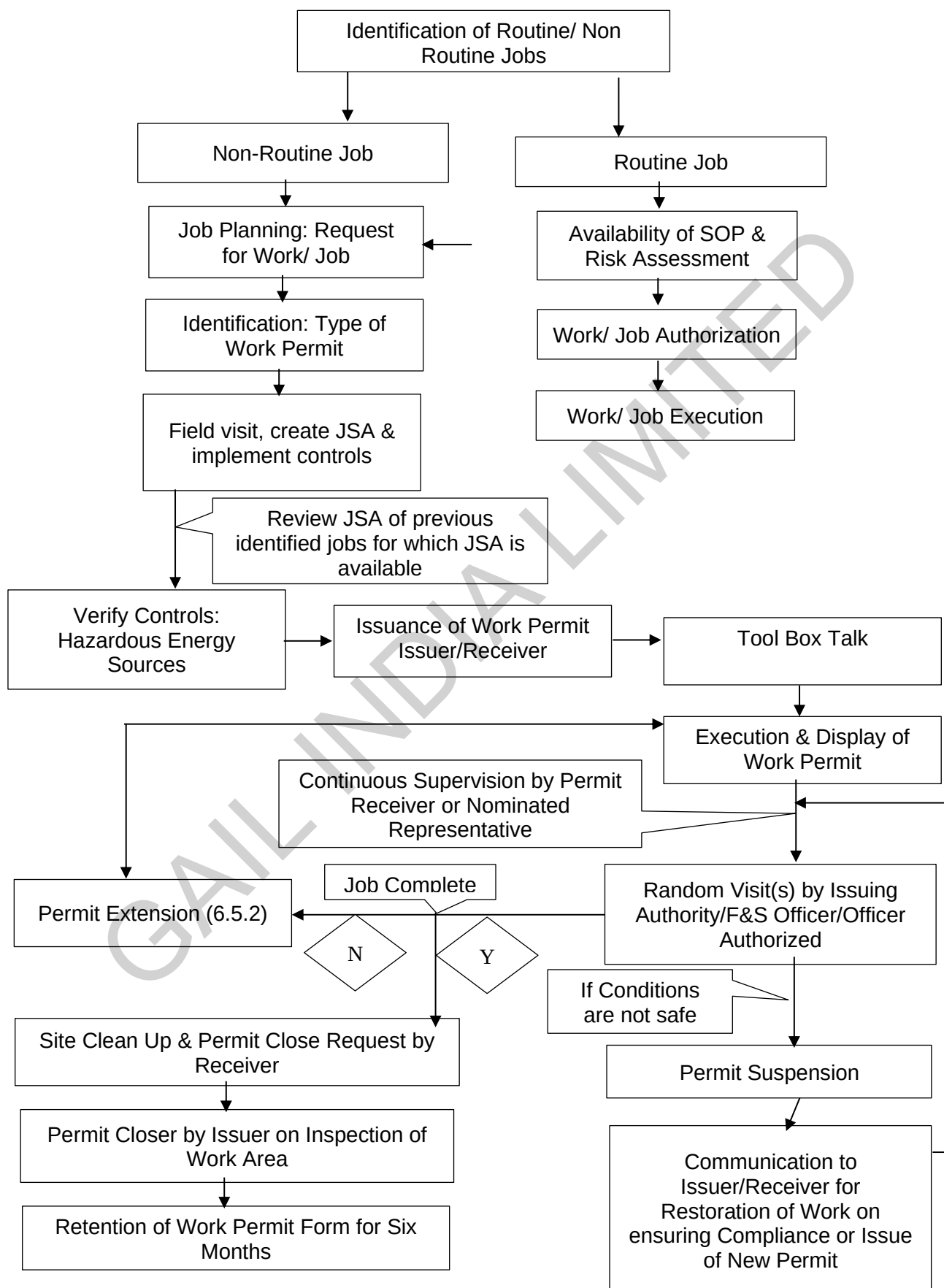


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ANNEXURE – IV





ANNEXURE- V

SAFE WORK PRACTICES

Organizations shall develop safe work practices to provide for the control of hazards. The practices shall support a work permit system in managing these activities safely. The practices shall include following minimum requirements to execute these activities to prevent potential major events including release of flammable or toxic material or serious injury to the people.

1.0 HOT WORK

Following the seven key principles along with other good safety practices which can prevent deaths and injuries from hot work:-

- (a) Use Alternatives – Whenever possible, avoid hot work and consider alternative methods.
- (b) Analyze the Hazards – Prior to the initiation of hot work, perform a hazard assessment that identifies the scope of the work, potential hazards, and methods of hazard control.
- (c) Monitor the Atmosphere – Conduct effective gas monitoring in the work area using a properly calibrated combustible gas detector prior to and during hot work activities, even in areas where a flammable atmosphere is not anticipated.
- (d) Test the Area – In work areas where flammable liquids and gases are stored or handled, drain and/or purge all equipment and piping before hot work is conducted.
- (e) When welding on or in the vicinity of storage tanks and other containers, properly test and if necessary, continuously monitor all surrounding tanks or adjacent spaces (not just the tank or container being worked on) for the presence of flammables and eliminate potential sources of flammables.
- (f) Use JSA – Ensure that qualified personnel familiar with the specific site hazards review, develop JSA and authorize all hot work and issue permits specifically identifying the work to be conducted and the required precautions.
- (g) Train Thoroughly – Train personnel on hot work policies/procedures, proper use and calibration of combustible gas detectors, safety equipment, and job specific hazards and controls in a language understood by the work crew.
- (h) Supervise Contractors – Provide safety supervision for outside contractors conducting hot work.
- (i) Inform contractors about site-specific hazards including the presence of flammable materials.

Other Precautions to be taken while planning and executing hot work are:-

- (a) Welding machine shall be kept in a safer location and ventilated area and shall be properly earthed. Before commencing the welding job, ensure that all the cables and earthing connections are as per the standard welding codes / practices. It shall be switched off immediately on completion of the job.
- (b) While performing welding / cutting jobs in confined spaces, ensure that cylinders (oxygen, acetylene, nitrogen, argon, etc.) are kept outside separately and hoses for the gases are in good working condition. Flash back arrestor shall be provided for cutting the torch.
- (c) Gas tests for hydrocarbons, oxygen deficiency, toxic gases, etc. shall be conducted as applicable as a prerequisite to issue permits for hot work/ vessel entry etc. No hot work shall be permitted unless the Explosive meter reading is zero.
- (d) The gas testing shall be conducted as necessary to assure there is no accumulation of hazardous vapours. The atmosphere shall be tested either continuously or for a maximum interval of two hours.
- (e) If the level of LEL above the prescribed limit is detected, the gas source shall be identified, and the leakage shall be rectified. The job shall be suspended during this time period. The permit conditions shall be reviewed. In case the original permit conditions are not restored, the permit shall be closed.

Refer OISD-GDN-192 & IS 3016 for safe practices to be followed while cutting/ welding.



2.0 WORKING AT HEIGHT

Work at height means work in any place where, if there were no precautions in place, a person could fall a distance liable to cause personal injury. For example, person/s are working at height if they:

- (a) Are working on a ladder or a flat roof.
- (b) Could fall through a fragile surface.
- (c) Could fall into an opening in a floor or a hole in the ground.

Before working at height, one shall work through these simple steps:

- (a) Avoid work at height where it is reasonably practicable to do so.
- (b) Where work at height cannot be avoided, prevent falls using either an existing place of work that is already safe or the right type of equipment.
- (c) Minimize the distance and consequences of a fall, by using the right type of equipment where the risk cannot be eliminated.
- (d) Ensure workers can get safely to and from where they work at height.
- (e) Ensure equipment is suitable, stable and strong enough for the job, maintained and checked regularly.
- (f) Make sure you don't overload or overreach when working at height.
- (g) Take precautions when working on or near fragile surfaces.
- (h) Provide protection from falling objects.
- (i) Consider emergency evacuation and rescue procedures while planning the job.

Refer OISD-GDN-192 for Work at height

3.0 CONFINED SPACES

- (a) No person shall be required or allowed to enter any confined space, until all practicable measures have been taken to remove any gas, fume, vapour or dust, which may be present so that to bring its level within the permissible limits and to prevent any ingress of such gas, fumes, vapours or dust and unless; Entry permit in writing has been issued based on the test carried out that the space is reasonably free from dangerous gas, fume, vapour or dust.
- (b) Vessel entry, where no hot work is to be carried out, may be permitted under extra precautionary measures based on the careful analysis of the potential hazards if combustible gases are up-to 5% of lower explosive limit (LEL). Entry with an air-supplied mask may be permitted with LEL of up to 20%. The oxygen level shall be at least 19.5 % vol. and the concentration of toxic gases below the threshold limits.
- (c) The person entering the confined space shall be equipped with a personal gas detector. In addition, the standby person or Safety watch shall have a portable gas detector to monitor the confined space environment periodically. The gas detector shall be calibrated every 6 months / OEM instructions and its functioning shall be verified before use. The records of persons entering and exiting the confined space shall be maintained.
- (d) Special attention shall be given for confined space entry into vessels fitted with nucleonic gauges for level measurement which use radioactive elements (Cesium 137 or cobalt). The radioactive source holder is typically a lead block with a small aperture and shutter encased in a carbon steel enclosure. During maintenance and inspection, such vessels need to be inspected by entering the vessel. Before entry, it must be ensured that the source is isolated and permit issuer shall specifically mention the same in confined space entry permit.

4.0 HEAVY LIFTS

Typical incidents have included: the failure of lifting equipment; inappropriate lifting and slinging practices; inadequate supports or supports not resting on level/firm ground; incorrectly estimating of the weight or center of gravity of the load, and over estimation of an individual's capability to support a load or restrain its movement.



If a heavy item has to be moved or temporarily supported during maintenance work, it is crucial that the risks involved are assessed and a plan of action is properly thought through.

Following precautions must be taken while planning Heavy lifts: -

- (a) Carry out JSA involving rigging experts who can provide advice on safe slinging and on safe working practices for work involving heavy loads.
- (b) Decide plan of action before a heavy item is moved or temporarily supported.
- (c) Ensure the rigging plan is prepared by the qualified and authorized person.
- (d) Make sure any equipment used to lift or support a heavy load is suitable and (where necessary) has had its inspection and test by a competent person.
- (e) Do not leave heavy items unsecured where they may tip over, fall or slip, and don't work under suspended loads that haven't been properly propped in position.
- (f) Don't overestimate the ability of a person to prevent a heavy item from moving.

5.0 EXCAVATION WORK

For excavation work regardless of the depth, permission from various sections shall be obtained with precautions to be taken for the underground facilities viz; sewers, telephone lines, cables, pipelines etc. Copy of permission for excavation shall be attached to the work permit. A feedback system shall be in place to inform all the concerned departments/ sections about the completion of work.

Following hazards of excavation shall be considered while planning the job:-

- (a) The collapse of the sides of the excavation.
- (b) Materials falling onto people.
- (c) Falls, either people or vehicles.
- (d) Nearby structures collapsing into the excavation.
- (e) Electrocution, explosion, gas leak, or flooding, caused by damage to underground services.

Refer OISD-GDN-192 for safe practices during excavation.

6.0 VEHICLE ENTRY

Vehicle Entry permits are used whenever vehicles (including heavy duty equipment such as cranes, front end loaders, back-hoes) will be entering areas where there is:

- (a) A possibility of volatile gas releases.
- (b) The potential for serious, major or critical accidents occurring because of the vehicle entry into the area (e.g. potential for the vehicle striking and damaging equipment).

7.0 LINE BREAK

One of the dangerous and hazardous tasks in the realm of both process safety and safety at all facilities is non-routine process openings, commonly called line breaks. The risks involved in these processes affect the employees performing the task, workers and contractors throughout the entire site, and potentially could even impact off site receptors in the form of people and the environment.

Following precautions shall be considered while planning and executing the Line Breaks: -

- (a) The exact location of the break in must be identified prior to work commencing and the undertaking of a risk assessment.
- (b) For ALL line breaks, obtain and read the chemical SDS before performing any line break or work where the hazard of chemical exposure exists. Make sure appropriate PPEs are identified and worn as determined by the chemical manufacturer's SDS.
- (c) Prior to beginning the work, verify operability of the closest safety shower/eye wash station.



- (d) Utilize required Personal Protective Equipment on the initial break into an isolated section and any subsequent breaks when it could be reasonable to expect trapped material, such as in the case of a low point in the line.
- (e) Ensure PPE is in good condition and no noticeable defects that would expose you to the hazard
- (f) Treat all lines as if they are under pressure. Break flanges "down and away" (out of the line of fire). The initial break shall be made over containment whenever possible. Be prepared to re-tighten bolts if the line is not clear.
- (g) When breaking into an isolated section, the discovery or suspicion of plugged or pressurized equipment or piping requires that work be stopped, and operations personnel be consulted to develop a plan for unplugging and de-pressurizing.
- (h) PPE requirements may be downgraded when it has been verified that the risk has been eliminated.

8.0 ROAD WORK AND ROAD CLOSURE

For any repair work or work involving closure of one or more roads inside the facility, the road closure permit procedure shall be followed:

- (a) A road closure permit is required for a job which requires blocking of a road, partial or full, which can restrict the movement of emergency vehicles.
- (b) No Road closure shall be permitted unless alternate routes are identified and accepted by emergency services.
- (c) Necessary signage shall be displayed at such prominent locations so that vehicles will not have to go back to find alternate route.
- (d) A Minimum 4 m width of the road shall be available for consideration of partial road closure.
- (e) All jobs required road closure shall be planned with necessary approvals from all stakeholders with communication to all.
- (f) Work Permits which are issued for carrying out the job during the road closure shall be linked on to the composite work permit.
- (g) Single point responsibility shall be assigned to suspend / terminate a road closure permit in case of emergency.

9.0 DIVING

Diving is a high hazard activity. Diving certificate can be used to control the diving activity itself and to ensure that there are no other activities taking place nearby which create unnecessary additional risks.

Diving safety is the aspect of underwater diving operations and activities concerned with the safety of the participants. The safety of underwater diving depends on four factors: the environment, the equipment, behavior of the individual diver and performance of the dive team.

There is a large range of hazards to which the diver may be exposed. These each have associated consequences and risks, which shall be considered during dive planning. Where risks are marginally acceptable it may be possible to mitigate the consequences by setting contingency and emergency plans in place, so that damage can be minimized were reasonably practicable.

The JSA during job planning shall address all the hazards before the permit is prepared for diving activity.

10.0 RADIOGRAPHY/ WORK ON RADIATION SOURCE

Radiography with an X-ray machine shall be carried out under hot work permit. Cold work permit may be used if it is done with radio-active isotopes. Radiography shall be carried out when occupancy is least. Based on the maximum allowable radiation, areas shall be cordoned off. Adequate warning signs on the outskirts of the cordoned area shall be provided. Warning lights shall be used for night radiography. Refer OISD-GDN-192 & guidelines prescribed by Atomic Energy Regulatory Board (AERB) for industrial radiography practice.



Nuclear gauges have been used in a wide range of industries to improve the quality of products, optimize processes, and save energy and materials. Nuclear gauges have a radioactive source that is covered by a radiation blocking shield. Gauges can contain a gamma, beta or neutron radiation source. Different sources are used depending on what the nuclear gauge is measuring.

If a radioactive source is present on or nearby any equipment where maintenance work is to be performed, the radioactive source has to be isolated before such a work is started. If isolation is not possible, the source must be physically removed.

The radioactive source Isolation and restoration shall be done by issuing separate work permits.

11.0 HIGH PRESSURE WATER CLEANING

Water jetting is vital to the Oil & Gas, chemical and petrochemical industry. It is used for cleaning and de-fouling a huge range of equipment, including heat exchangers, process vessels, holding tanks, large-diameter pipes and site drainage systems.

It is also used for surface preparation during construction and structural maintenance and intrinsically safe cold-cutting. The major hazards include:-

- (a) Injury due to high pressure water jet.
- (b) High-velocity impact.
- (c) Chemical Exposure.
- (d) Musculoskeletal Injury.
- (e) Machinery and equipment.

Following precautions shall be considered while planning and executing the High-Pressure water cleaning tasks: -

- (a) The area around the job, pump, and hoses shall be barricaded. The minimum boundary shall be decided by the engineer in charge depending upon the location, area, density of people around etc. and signs shall be displayed stating "DANGER - HIGH-PRESSURE WATER CLEANING" at the perimeters.
- (b) Hose with exposed or damaged wire braid will be removed from high-pressure service.

12.0 ONLINE LEAK SEALING

Repair clamps are sometimes used as a temporary repair to seal small leaks in pipework. There are instances of failure of pipelines while executing the online sealing causing fatalities.

Following precautions shall help prevent such incidents: -

- (a) The use of leak sealing clamps shall be adequately controlled by appropriate procedures. Users of leak sealing repair clamps shall carry out the following steps prior to the use of such a clamp:
 - (i) An assessment of the nature and extent of the defect being considered for repair.
 - (ii) A full assessment of the operating conditions to which the clamp will be subjected. This shall include pipe contents, pressure, temperature, pipe size and combinations thereof.
 - (iii) An assessment of the timescale for which the clamp is required to be in operation.
- (b) If a leak sealing repair clamp is considered suitable following these assessments, then this information shall be communicated to the clamp supplier.
- (c) It shall be ensured that all pipes enclosed by repair clamps are provided with structural restraint unless a suitable and sufficient assessment. It is important to consider the ability of the remaining wall thickness to accommodate the imposed axial loadings/stresses. In situations where the continuation of internal metal loss cannot be prevented it will be necessary to consider the suitability of the repair component to accommodate these loading/stresses shall this continuing metal loss affect the pipe's ability to carry these loading/stresses. Additional measures to provide sufficient structural restraint for axial loads shall be available from clamp suppliers.



- (d) Great caution shall be taken where there is a need to re-inject existing clamps and structural restraint has not been provided. A suitable and sufficient JSA of the conditions in which the clamp has been employed shall be carried out.
- (e) Leak sealing repair clamps are temporary repairs. Such temporary repairs shall be removed and replaced with a suitable permanent repair at the next available maintenance opportunity. Temporary repairs may remain in place for a longer period of time only if approved and documented by a person competent to do so.

13.0 HOT TAPPING

Hot-tap is a connection technique to take new branches from the existing pipeline which is under pressure or in service. A special procedure is required to make a branch on the pipe without leaking any of the line contents.

Even though there are API codes and guidelines available for conducting welding/drilling on live systems, each facility is unique. It is recommended that each Organization shall develop an exhaustive procedure for carrying out hot tapping on live equipment at their premises.

The procedure shall detail the technical requirements and the responsibilities of the various agencies/personnel involved in the execution of the job.

The procedure shall indicate fluids used at the facility for which hot tapping is prohibited. As an example, fluids that can cause stress corrosion cracking in metals shall not be considered for hot tapping.

Considering the serious hazards, a well-documented approval matrix shall be put in place for authorization of each of the hot tapping jobs.



ANNEXURE – VI

TYPICAL ROLES & RESPONSIBILITIES OF WORK PERMIT ROLE HOLDERS

The employer, site occupier, or installation duty holder (i.e., the installation owner or operator) is responsible for ensuring proper work permit systems are developed and followed.

Following are the responsibilities and duties of role holders under a work permit system:

(1) Site or Installation Managers, shall ensure that:

- (i) All work requiring a work permit is identified.
- (ii) The permit contains a clear description of the work to be done, its location, start time and duration.
- (iii) Permits for work activities that may interact or affect other site activity are adequately controlled.
- (iv) All other work that would create a hazard if undertaken at the same time is suspended and made safe.
- (v) Limitations on the timing and scope of the work are defined as well as actions to be taken in the event of site emergencies.
- (vi) All personnel engaged in the preparation of permits and responsible for the supervision and performance of the work are identified and competent.
- (vii) Sufficient information is given to oncoming shifts about work for which there is a permit and which has not been completed.
- (viii) All personnel (including contractors) working within the permit system have sufficient knowledge and competence to carry out their duties.

A responsible person (referred to as the 'area authority') shall be appointed to co-ordinate and control the issue and return of permits. That person shall have an overview of all operations under way and planned on site to avoid hazards caused by simultaneous activities.

The Site or Installation Manager is normally responsible for ensuring this coordination and control, either by controlling the issue and return of permits themselves (a preferable arrangement on smaller offshore platforms or where there are only a small number of permits daily), or by appointing an appropriate responsible person (or people) with sufficient authority to carry out this function on their behalf.

(2) Permit issuers, shall ensure that:

- (i) All hazards associated with the proposed job have been identified and suitably assessed.
- (ii) The work site has been examined, and all precautions specified to be taken before work commences (including isolations) have been taken and will remain effective while the permit remains in force.
- (iii) The permit receiver is aware of the precautions taken, any additional ones which are to be taken, particular equipment to be used or worn, and any other procedures which are to be followed.
- (iv) Work activities that may interact or affect one another are identified, and either conflict avoided or precautions included on the permit (e.g., use of welding shields).
- (v) People are aware of the permit's duration and action to be taken if the work is suspended.
- (vi) Copies of all issued permits are displayed at an appropriate location and in a consistent arrangement so that site personnel can readily see and check which equipment is under maintenance and not available for operation.
- (vii) The work site is examined at any time when work is suspended and before it is restarted, and finally when the work is completed to ensure that it is in a safe condition.
- (viii) The shift handover procedure is followed correctly.
- (ix) Any precautions and isolations are withdrawn at the end of the job unless they are cross-referenced to other permit activity.
- (x) The area authority has acknowledged the return of plant or equipment to their full control.

The Occupier/ Factory Manager/ Installation/ Site Manager may be the Permit Issuer or delegate the permit issuing authority to a responsible, competent person the Organization authorizes.



It is essential that people authorized to issue permits-to-work have sufficient knowledge about the hazards associated with the relevant plant, to allow them to identify those hazards and control measures (e.g. isolations) correctly. If authorized people are relocated to former workstations, then refresher training shall be given and recorded before they are reauthorized.

(3) Permit receivers, shall ensure that:

- (i) They understand the operation of (and the consequences of non-compliance with) the permit-to-work systems applicable to the areas in which they are responsible for work.
- (ii) The conditions and precautions specified in the permits are fully understood, implemented and effectively monitored.
- (iii) The issuing authority is informed when a job has been completed, suspended, if conditions alter or if the task needs to be altered.
- (iv) They do not start work on any job requiring a permit until one has been authorized and issued, its content understood and necessary precautions taken.
- (v) If in any doubt, or if any circumstances or conditions change, they stop work, make the work area safe and get advice immediately.

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ANNEXURE - VII

TYPICAL FORMAT OF JOB SAFETY ANALYSIS (JSA) FORM

(NAME OF ORGANIZATION)

(LOCATION)

Date:	Job Location:	Job type:	Work Permit & isolation certificates Ref. No.:	
Job Executed by: (give firm name & PO number)		Job Description:		
JSA prepared by:		Name	Designation	Sign with date
1.				
2.				
3.				
4.				
JSA reviewed & approved by:				
1.				
2.				
S. No.	Job steps	Potential hazards	Hazard control measures	Person Responsible
Additional comments (if any):				



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ANNEXURE - VIII

LOCK OUT & TAG OUT PROCEDURE OF ELECTRICAL INSTALLATIONS

The term 'lock out' or 'tag out' (LOTO) refers to a procedure aimed at safeguarding someone working on or near a plant or machinery from that plant or machinery unexpectedly starting up or releasing energy of some kind. This procedure involves an authorized person disconnecting the plant or machinery from its energy source. This person then locks or tags the isolator to prevent anyone from re-energizing the plant or machine.

A lock-out device is a physical restraint that maintains the isolation device in a safe or 'off' position. It can only be removed with a key, or some other unlocking mechanism, which can only be used with the authority of the person in charge of the work permit system. For safety reasons, each lock must only have one key. No master keys are allowed on a lock-out system. Figure 3.0 shows an example of a lock-out device fitted to an isolator that prevents the isolator from being switched on.



Fig-3: Typical example of a Lock-out device

A tag-out device is a warning system that an authorized worker applies to the isolator to indicate work is being carried out on the plant or machine. As such, the isolator shall not be switched on while the work is underway. Tag-out devices are easy to remove. Consequently, they offer less protection than lock-out devices.



Fig-4: Typical example of a Tag out device

The general procedure for implementing lock-out and tag-out devices is as follows:-

- (i) The person authorized to conduct the lock-out or tag-out action will identify the source(s) of energy to be controlled and the control method to use. A safe working procedure will then be established to implement the isolation, followed by the lock-out and tag-out action.
- (ii) All personnel who may be affected by the isolation caused by the lock-out and tag-out action shall be informed. This information will include:
 - (i) What plant/machinery is going to be isolated?
 - (ii) Why it is going to be isolated.



- (iii) How long it will be isolated.
 - (iv) Who is the authorizing person?
 - (v) Who is the contact person?
- (iii) The process or system shall be shut down as normal, and confirmation made that all controls are in the off position and that all moving parts have stopped.
- (iv) The process or system shall be isolated from its energy source, following the safe working procedure set out by the authorized person at the identification of the energy stage. Depending on the source of energy, the procedure will vary as follows: -
- (i) **Electrical energy** – the isolator to be switched to the off position and the breaker connections to be visually verified to be in the off position. The isolator switch is locked in the off position with a lock-out or tag-out device.
 - (ii) **Hydraulic or pneumatic energy** – the isolator valve shall be set in the closed position and locked in the closed position with a lock-out or tag-out device. Pressure shall be bled off by opening the pressure relief valve.
 - (iii) **Mechanical energy** – energy shall either be carefully and slowly released from compressed spring, or the components shall be blocked in the system or process which has the potential to be moved or activated by the spring's energy. If the components are blocked from moving, the block shall be locked in position with a lock-out or tag-out device.
 - (iv) **Gravitational energy** – an isolation block or pin shall be positioned to stop the system or process from moving. The block or pin shall be locked in position with a lock-out or tag-out device.
 - (v) **Chemical energy** – the isolator valve on the supply lines shall be set in the closed position. The isolator valve shall then be locked in the closed position with a lock-out or tag-out device. Where possible or appropriate, chemicals shall be bled off by opening the bleed valve.
- (v) Verification that the isolation is effective needs to be made. This is normally done by attempting to try to restart the system or process. If it does not start, the isolation has been effective. Once isolation is verified, the start-up controls must be returned to their off or neutral position.
- (vi) Once the work is completed, the lock-out and tag-out devices will need to be removed. This operation shall include:
- (i) Ensuring all tools and equipment have been removed.
 - (ii) Ensuring all personnel are not near any hazardous areas.
 - (iii) All controls are in the off or neutral position.
 - (iv) Removal of the lock-out, tag out devices and re-energizing of the plant or process.
 - (v) Informing relevant personnel that the plant or process is back in service.
 - (vi) The person who has been undertaking the maintenance work shall remain while the plant or process is restarted.
- (vii) Temporary Removal of Isolations: -
- There are situations where isolations must be temporarily removed (e.g. when testing equipment during the work). When isolations must be temporarily removed, the work permit issuer must be informed and checks must be made to ensure that the isolation removal does not create any hazards for the work site and workers before the isolations are removed. After the requirement for the temporary isolation removal has been completed, the isolations should be reinstated and the Permit Issuer must be informed that the isolation's status.



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ANNEXURE - IX

TYPICAL FORMAT FOR TOOLBOX TALK FORM

(NAME OF ORGANIZATION)

(LOCATION)

DATE WITH TIME _____

WORK INSTALLATION/ UNIT/ FACILITY NAME _____

WORK PERMIT NO _____

CONTRACTOR NAME _____

TBT DELIVERED BY _____

CONTRACTOR SUPERVISOR NAME _____

TOPICS/ ISSUES DISCUSSED
OTHER POINTS OR ISSUES RAISED BY PARTICIPANTS

DETAILS OF PARTICIPANTS TO WHOM TBT WAS DELIVERED

NAME IN FULL	SIGNATURE	NAME IN FULL	SIGNATURE
1.		2.	
3.		4.	
5.		6.	
7.		8.	
9.		10.	



ANNEXURE - X

MAJOR CHANGES IN THE REVISED EDITION

1. The scope of the revised standard has been extended to the construction activities in Grass root/ Green field project sites.
2. The definitions of cold work, hot work, and authorized electrical person involved in electrical isolation/ energization of equipment in the revised standard have been updated in line with the Statutory Acts/ Rules/ Regulations.
3. Electronic work permit system is introduced in the revised standard for ease of business. The salient guidelines are:
 - (a) Digital signature of the Permit Issuer and Receiver is permitted in the work permit forms.
 - (b) The digital copy of the work permit form is permitted at the work site.
4. Any work at height more than 1.8 m is considered as "working at height" in the revised standard.
5. The detailed procedure for Job Safety Analysis (JSA), and Toolbox talk/ meetings (TBT) are incorporated in the revised standard. The typical formats of JSA and TBT forms are prescribed in Annexures.
6. The lockout and tagout procedure for energy sources has been incorporated in the revised standard.
7. The revised standard clearly specifies that the work permits shall not be issued and received by the same person.
8. In the revised standard, the permit issuing authority shall not be delegated to the contractor/ his representative except for two cases:
 - (a) Retail Outlets
 - (b) Installations where operations are outsourced on a contract/ turnkey project basis
9. In the revised standard, the permit receiving authority shall be the organization's authorized person. However, in some instances (like single-man locations), the permit may be directly received by the contractor's authorized supervisor. Such exceptions shall be documented in the organization's work permit system.
10. In the revised standard, the procedure of colour coding and distribution of copies of the work permit forms have been left to the Organization. The standard only gives a suggestive table.



ANNEXURE - XI

LIST OF PREVIOUS FUNCTIONAL COMMITTEES

1st FUNCTIONAL COMMITTEE
(February 1988)
LEADER

T.C. Varghese Hindustan Petroleum Corporation Limited

MEMBERS

V.S. Save Hindustan Petroleum Corporation Limited

G. Raghunathan Hindustan Petroleum Corporation Limited

S.V. Puthli Hindustan Petroleum Corporation Limited

N. Lal Oil & Natural Gas Corporation Limited

V.K. Ramesh Oil & Natural Gas Corporation Limited

M.N. Gogoi Oil India Limited

M.A. Sreekumar Cochin Refineries Limited

CO-OPTED MEMBERS

A. Sinha Oil India Limited

A. Varadarajan Madras Refineries Limited

MEMBER COORDINATOR

B.K. Trehan Oil Industry Safety Directorate

2nd FUNCTIONAL COMMITTEE
(July 1998)
LEADER

T.K. Anjaneyulu Hindustan Petroleum Corporation Limited (Refinery)

MEMBERS

R.N. Sharma Indian Oil Corporation Limited (R&P Division)

T.S. Raghavan Madras Refineries Limited

E. Nandakumar Cochin Refineries Limited

K.V. Seshadri Bharat Petroleum Corporation Limited (Refinery)

K.V. Moorthy Oil & Natural Gas Corporation Limited

A.K. Sharma Bongaigaon Refinery & Petrochemicals Limited

MEMBER COORDINATOR

S.N. Mathur Oil Industry Safety Directorate



3rd FUNCTIONAL COMMITTEE (September 2004)

LEADER

S.P. Garg GAIL (India) Limited, Pata.

MEMBERS

V. V. Kurup	Kochi Refineries Limited, Kochi
K.T.Lokhande	Deptt of Explosives, Nagpur
C.Bose	Oil India Limited, Duliajan
R.Abdul Raheem	Chennai Petroleum Corporation Limited, Chennai
S. M. Singh	Indian Oil Corporation Limited, Pipelines Division, Noida
A.K.Singh	Indian Oil Corp. Limited, Guwahati Refinery
Suhas Kate	Hindustan Petroleum Corp. Limited, Visakh Refinery
A.Tilwankar	Bharat Petroleum Corporation limited, Mumbai
M. K. Manocha	Oil and Natural Gas Corp., Mumbai.
A. H. Karande	Bharat Petroleum Corp. limited, Mumbai Refinery

MEMBER COORDINATOR

S. K. Aggarwal Oil Industry Safety Directorate, New Delhi

(In addition to the above, several other experts from the industry contributed to the preparation, review and finalization of this document.)